Università di Pisa

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Alessandro Gori

Relatore:
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Outline

1. Cuprates and Hubbard physics
2. The Hartree-Fock method
3. The normal phase
4. The antiferromagnetic phase
5. The superconducting phase
6. Conclusions and outlook

Cuprates and Hubbard physics

The timeline of superconductivity

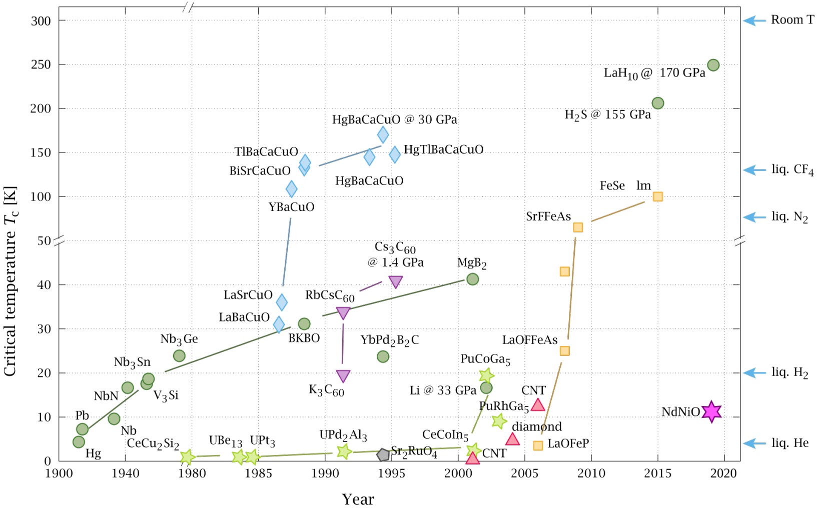
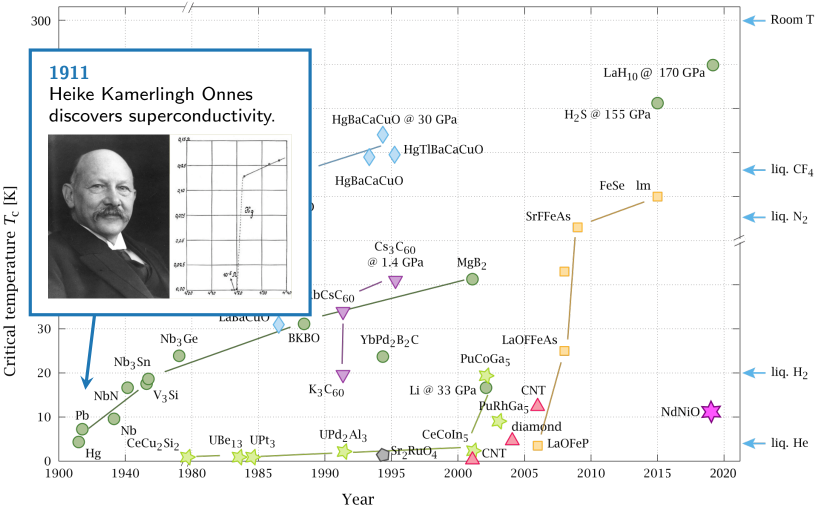


Figure 1: Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to the license CC BY-SA 4.0.

The timeline of superconductivity



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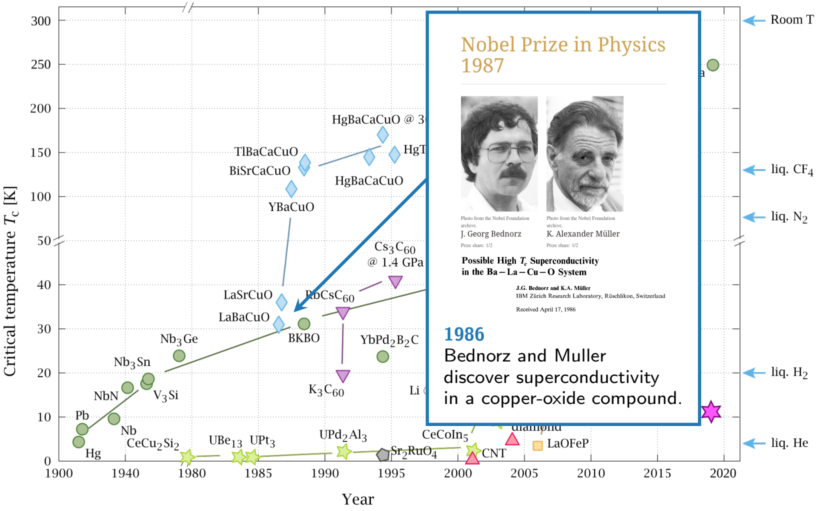

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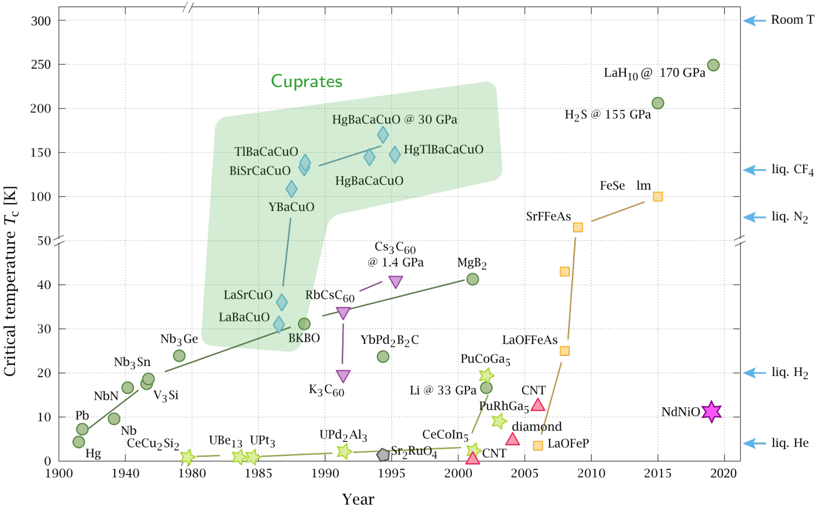


Figure 1: Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to the license CC BY-SA 4.0.

The phase diagram of cuprates

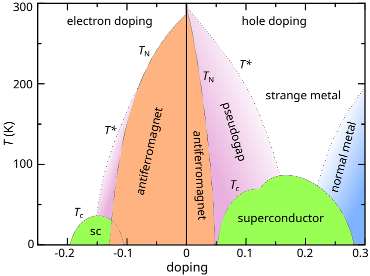
Cuprates are the prototypical example of **strongly correlated materials**:

- **Antiferromagnetism** at null and low doping;
- d-**wave superconducting** domes.

Both arise from **strong electronic correlations**.

$$\Downarrow$$

Failure of band theory in microscopic description.

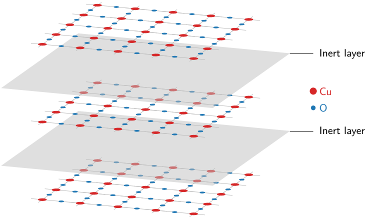


Schematic phase diagram of cuprates. Author: Holger Motzkau, Wikimedia commons, distributed according to the license CC BY-SA 4.0

Cuprates and the Hubbard model

These materials are stackings of:

- 2D copper oxide active layers;
- intermediate layers of inert material.



$\Downarrow$

The single-band Hubbard model on the square lattice

$$\hat{H}_{\mathrm{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \qquad \text{with } t, U > 0.$$

Despite its simplicity, the Hubbard model describes:

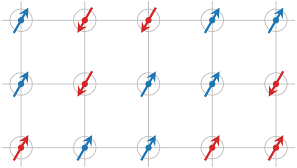
- Insulation at half-filling (**Mott localisation**);
- **Antiferromagnetism** induced by Mott localisation;
- d-**wave superconductivity** emerging by doping the antiferromagnetic phase.

What is the origin of the insulating state at half-filling?

The Hubbard model at strong coupling,

$$t \ll U ,$$

describes the **Mott insulator** state.



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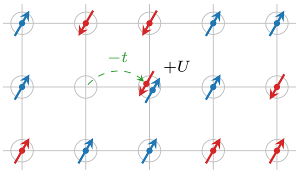

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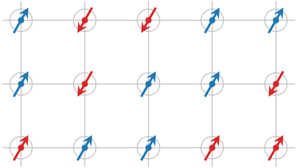
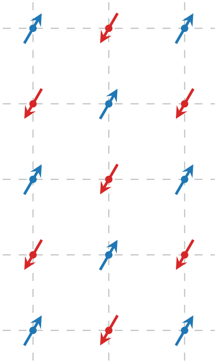
$$\Downarrow$$

Mott insulation mechanism

Electrons are **localised** as in a sort of "traffic jam".



What is the origin of the antiferromagnetic ordering?



Antiferromagnetism in the undoped Hubbard model

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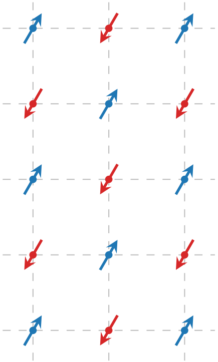
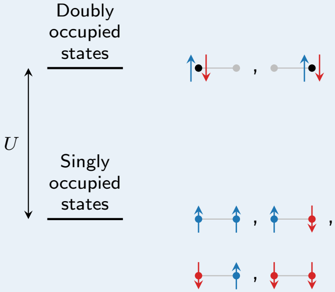


Example: 2 sites, 2 electrons

At $U \gg t$, singlet pairing is energetically favoured:



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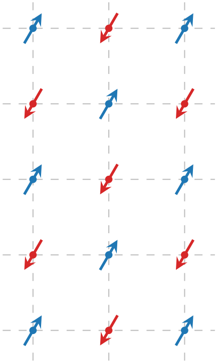

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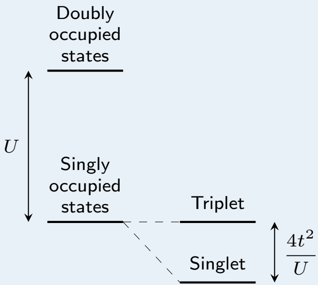

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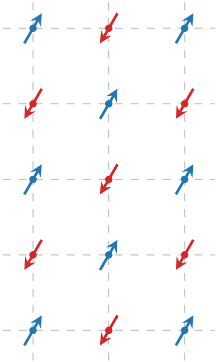

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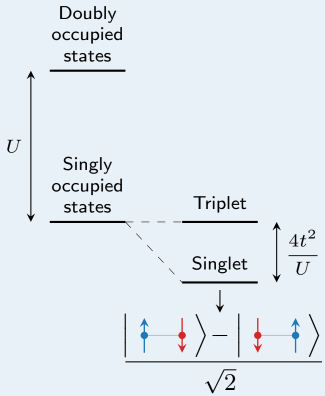
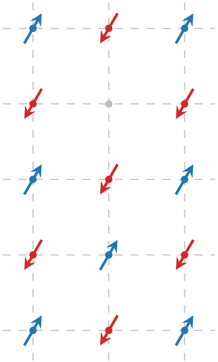


How does superconductivity emerge from doping the AF state in the Hubbard model?



Superconductivity emerges from attractive interaction between electrons.

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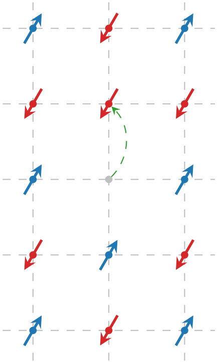

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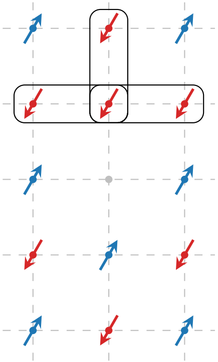

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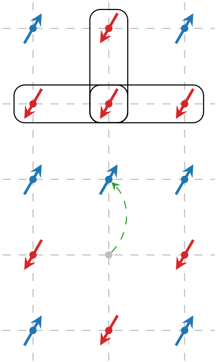

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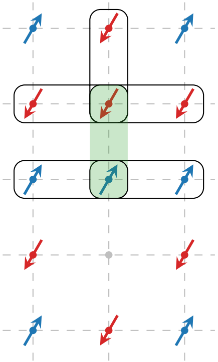

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Neighbouring electrons with opposite spin "move together":

$\Downarrow$

Effective attractive interaction...

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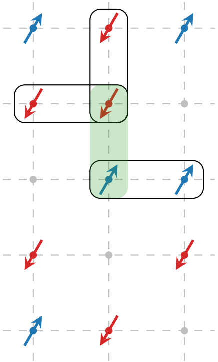


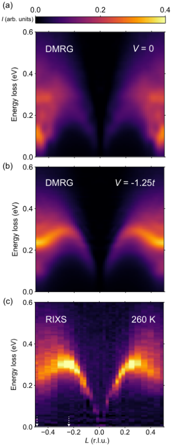
Superconductivity emerges from attractive interaction between electrons.

Neighbouring electrons with opposite spin "move together":

$$\Downarrow$$

Effective attractive interaction... favoured by doping.

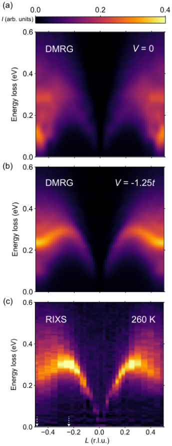
Recent developments: the extended Hubbard model



Hari Padma et al., 2025. **Physical Review X 15 (2): 021049**.

Observed: incompatibilities of the Hubbard model with the spectrum of magnetic excitation in cuprate $\mathrm{Sr_{14}Cu_{24}O_{41}}$.

Recent developments: the extended Hubbard model



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Observed: incompatibilities of the Hubbard model with the spectrum of magnetic excitation in cuprate $\mathrm{Sr_{14}Cu_{24}O_{41}}$.

$$\Downarrow$$

Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\mathrm{EHM}} = \hat{H}_{\mathrm{HM}} \underbrace{-V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V} .$$

$\hat{H}_V$ can act **as an independent source of** d**-wave superconductivity**.

The model (EHM):

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \qquad \text{with } t, U, V > 0.$$

The motivation: compatibility with some aspects of cuprates physics, including d-wave superconductivity.

The target: understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

The method: Hartree-Fock variational method, at low temperature.

The Hartree-Fock method

The constrained minimisation framework

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0.

Variational principle:

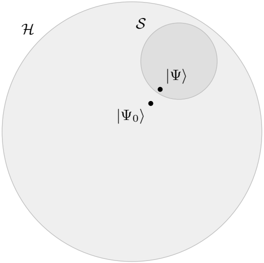
$$\forall\, |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} \quad : \quad E[\Phi] \geq E_0\,.$$

$$\Downarrow$$

Constrained minimisation procedure

$$|\Psi\rangle \quad : \quad \left.\frac{\delta E[\Phi]}{\delta\,|\Phi\rangle}\right|_{|\Phi\rangle = |\Psi\rangle} = 0\,.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

The Hartree-Fock subspace of states

$\mathcal{S}$ is the set of all **gaussian states**,

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}^{\dagger}_{\alpha} |\Omega_{\gamma}\rangle \ , \qquad \left\{\hat{\gamma}_{\alpha}, \hat{\gamma}^{\dagger}_{\beta}\right\} = \delta_{\alpha\beta} \ ,$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

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with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

$$\Downarrow$$

Each gaussian state can be identified **uniquely** by fixing

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \,, \qquad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \,.$$

These are the **variational parameters** collected in the vector $\mathbf{v}$. We aim to find the optimal $\mathbf{v}$, such that

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle \,.$$

Phase transition: spontaneous breaking of some symmetries of the model.

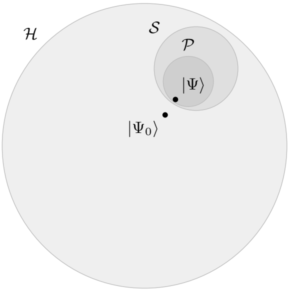
Phase X: shows symmetries Y, Z ...

$$\Downarrow$$

Restrict to $\mathcal{P} \subset \mathcal{S}$, gaussian states with symmetries Y, Z ...

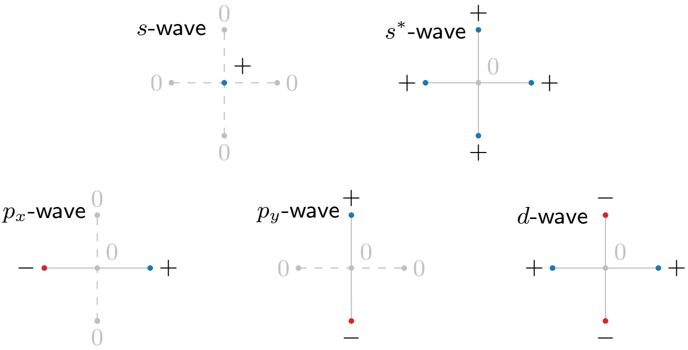
$$\Downarrow$$

Symmetry selection rules on $\mathbf{v}$ entries.



Identify symmetries $\rightarrow$ Derive $\mathbf{v}$ $\rightarrow$ Solve the variational problem.

It is useful to introduce five spatial symmetry channels:



Function f on NNs: decomposition in harmonics $f^{(\gamma)}$.

The normal phase

The variational problem for the normal phase

Normal phase: no broken symmetries. Just one variational parameter,

$$\mathbf{v} = \left[u^{(s^*)} \right] ,$$

where $u^{(s^*)}$ is the s^*-wave component of $u_{\mathbf{k}\sigma} \equiv \left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \right\rangle$.

$$\Downarrow$$

Renormalisation of kinetics

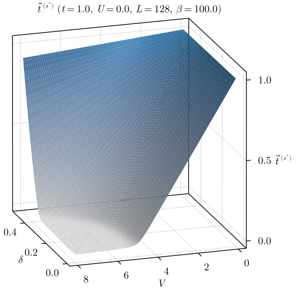
Hopping amplitude is **isotropically renormalised**,

$$t \quad \rightarrow \quad \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)} V}{2} ,$$

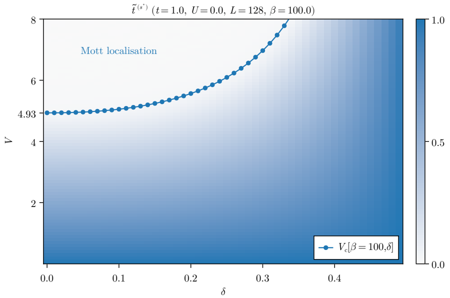
with the **theoretical constraint**:

$$0 < \tilde{t}^{(s^*)} < t \quad \Longrightarrow \quad \text{Possible Mott localisation.}$$

In (δ, V) plane, we have determined theoretically the presence of a Mott critical boundary $V_{\mathrm{c}}[\beta, \delta]$,



(a) Numerical results for $\tilde{t}^{(s^*)}$.



(b) Critical Mott boundary.

Within the Mott plateau, electrons are completely localised and **bands are flattened**.

The original results

The role of V/t:

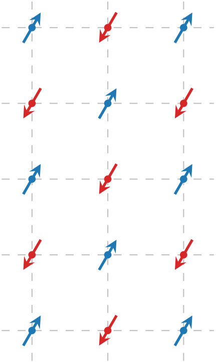
- **Renormalise hopping**;
- **Induce complete Mott localisation**:

$$\text{Effective kinetics energy scale} \ll \text{Interaction energy scale}\,.$$

The role of U/t:

- At HF level, U does not produce Mott localisation.

The antiferromagnetic phase



In cuprates, the AF phase is **commensurate**:

- 2-sites periodic in each direction;
- Sublattices oppositely magnetised;
- Zero net magnetisation.

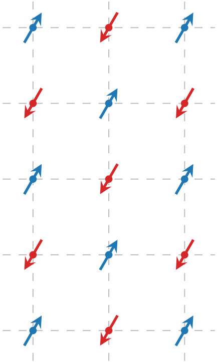
$$\Downarrow$$

Staggered magnetisation m:

$$2m = \left|\langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle\right|,$$

with i a site.

Commensurate antiferromagnetism in cuprates



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with i a site.

Half-filling restriction

We limit to $\delta = 0$ (half-filling) due to **stability considerations**.

The variational problem for the AF phase

Variational parameters for the AF phase:

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} ,$$

with $u^{(s^*)}$ defined as for the normal phase.

Renormalisation of kinetics

Mott localisation effect **formally** identical to the normal phase:

$$t \quad \rightarrow \quad \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)} V}{2} ,$$

with the **theoretical constraint**: $0 < \tilde{t}^{(s^*)} < t$.

The cooperation on magnetisation

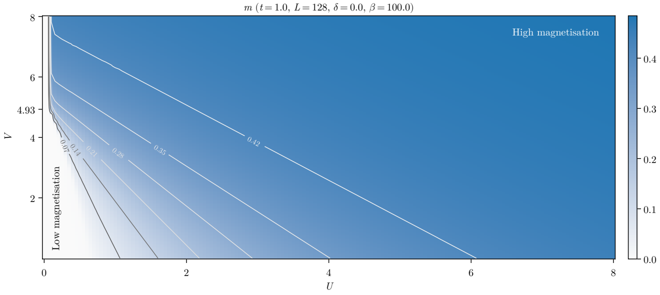
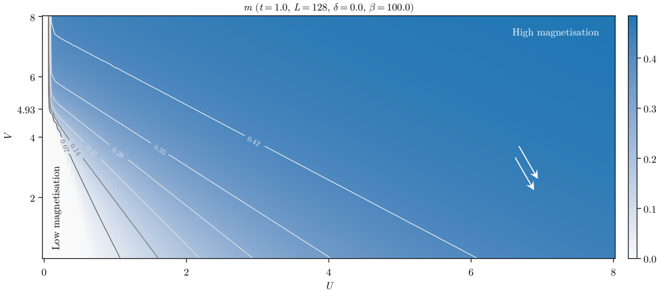


Figure 4: Numerical results for m. The gray curves indicate isovalued lines.

The cooperation on magnetisation



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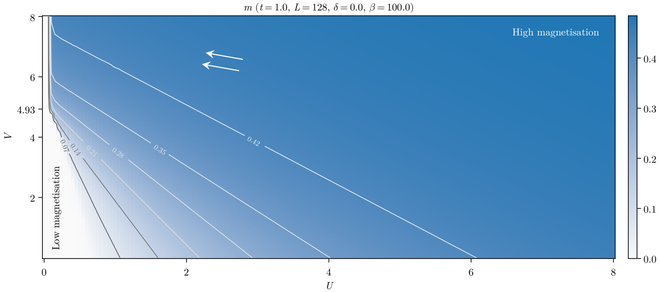
$U \gg V, t$

"Conventional" Hubbard magnetisation, induced by the ratio U/t.

The cooperation on magnetisation



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$V \gg U, t$

New "unconventional" source of magnetisation, induced by the ratio V/t.

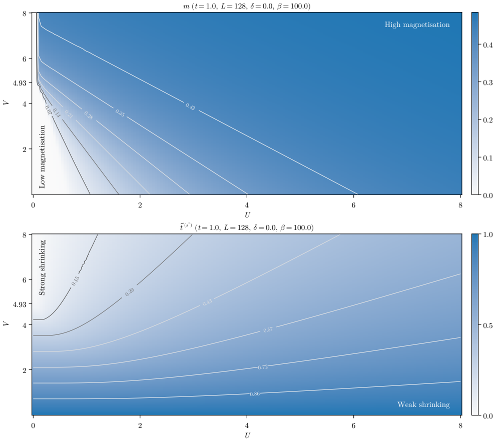
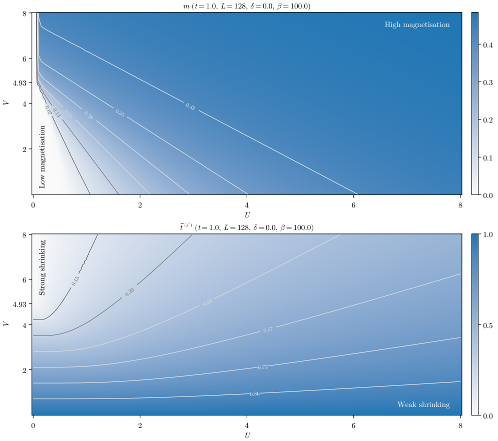


Figure 5: Numerical results for m (top) and $\tilde{t}^{(s^*)}$ (bottom). The gray curves indicate isovalued lines.

At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}\left(r^2\right) ,$$

where $r = \tilde{t}^{(s^*)}/U$.



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At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}\left(r^2\right) ,$$

where $r = \tilde{t}^{(s^*)}/U$.

$$\Downarrow$$

To enhance m, either:

- Increase U;
- **Decrease** $\tilde{t}^{(s^*)}$.

The original results

The role of V/t:

- **Renormalise hopping**;
- **Magnetise the lattice using Mott physics**;

The role of U/t:

- Magnetise the lattice, conventional Hubbard physics.

$$\Downarrow$$

Cooperation among interactions to magnetise the lattice.

The superconducting phase

The translationally invariant SC phase

Cooper pairing:

- Described by a SC gap $\tilde{\Delta}_{ij}$, roughly the "binding energy" of the pairs;
- The pair spin obeys sum rules:

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1 = \text{singlet} \oplus \text{triplet}.$$

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$$\Downarrow$$

The singlet channel

The gap function is inversion symmetric, i.e. its harmonics are

$$\tilde{\Delta}^{(s)}, \quad \tilde{\Delta}^{(s^*)} \quad \text{and} \quad \tilde{\Delta}^{(d)}.$$

We limit to the **singlet channel**. Let the quantities:

$$u_{\mathbf{k}\sigma} \equiv \left\langle \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \right\rangle , \quad w_{\mathbf{k}} \equiv \left\langle \hat{c}^{\dagger}_{\mathbf{k}\uparrow} \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \right\rangle .$$

Variational parameters: symmetric harmonics,

$$\mathbf{v} = \left(u^{(s^*)}, u^{(d)}, w^{(s)}, w^{(s^*)}, w^{(d)} \right)$$

The role of each parameter

- $u^{(s^*)}$, $u^{(d)}$: renormalisation of kinetics;
- $w^{(s)}$, $w^{(s^*)}$ and $w^{(d)}$: components of the SC gap.

Isotropic gap components:

$$\tilde{\Delta}^{(s^*)} = Vw^{(s^*)} \qquad \text{(Mediated by the isotropic non-local attraction)}$$

$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \qquad \text{(Mediated by the } \textbf{local repulsion}\text{)}$$

Anisotropic gap component:

$$\tilde{\Delta}^{(d)} = Vw^{(d)} \qquad \text{(Mediated by the isotropic non-local attraction)}$$

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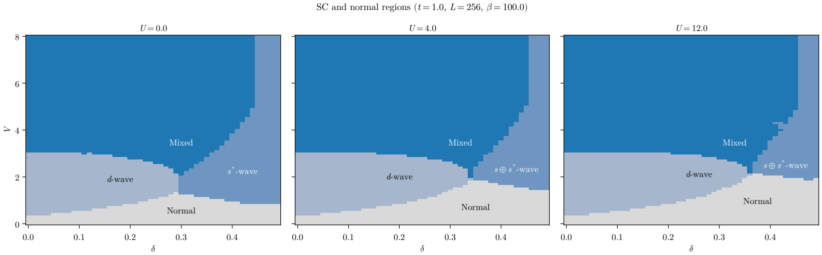

Figure 6: Phase diagram for the SC phase.

In the SC phase, the renormalisation of t gives two components,

$$t \quad \to \quad \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2}\,.$$

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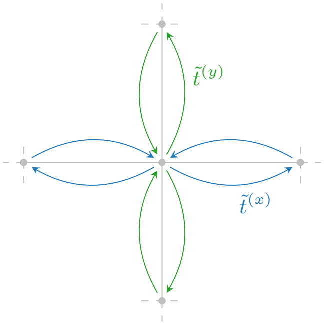


Anisotropic hopping

Direction-resolved effective hopping:

$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2} \, ,$$

$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2} \, .$$

$$\Downarrow$$

$\tilde{t}^{(d)} \neq 0 \implies$ **nematic SSB**.

Symmetry mixing and insurgence of nematicity

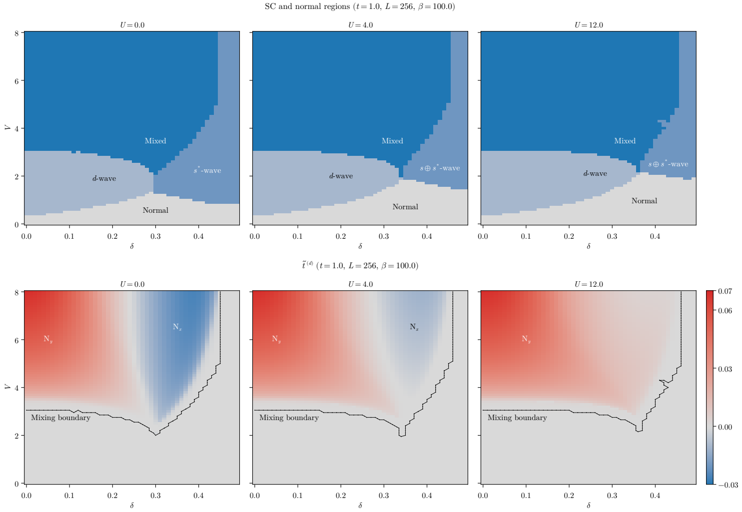


Figure 7: Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.

The role of V/t:

- **Renormalise hopping anisotropically**;
- **Induce mixed-symmetry superconductivity**;
- **Induce nematic symmetry breaking**.

The role of U/t:

- **Participate in superconductivity even if locally repulsive**.

Conclusions and outlook

Our original results, summarised

We wanted to understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

We conclude that at HF level, the non-local attraction:

- Acts as a source of Mott localisation;
- Enhances magnetisation at half filling;
- Can produce mixed-symmetry superconductivity;
- Can produce nematic superconductivity.

From here, we can follow various paths:

1. **Same model, other phases**.
   Triplet superconductivity, breaking of translational invariance, nematic antiferromagnetism...
2. **Expand the model**.
   Second nearest neighbours hopping, more refined potentials...
3. **More advanced numerical techniques**.
   DMRG for chains and ladders; hybrid DMFT (local sector) and HF (non-local sector) to retrieve Mott physics induced by U...

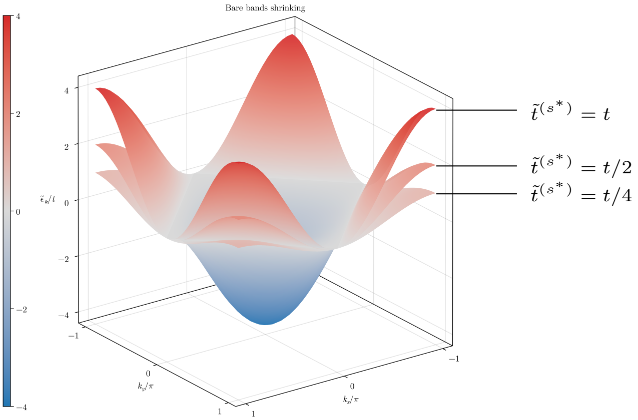
That's all, thank you!

Backup 1: details on Mott localisation

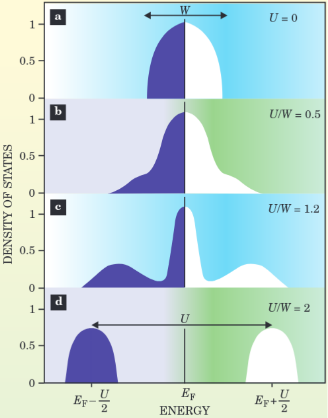
Bands flattening: a signature of Mott localisation (1/2)

Renormalisation of electronic bands, $\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma}$:

$$0 < \tilde{t}^{(s^*)} < t \quad \Longrightarrow \quad \textbf{Flattening} \text{ of bands} = \text{Mott localisation.}$$



Bands flattening: a signature of Mott localisation (2/2)



Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

As we increase the ratio U/t, the many-body DoS undergoes a transition:

(a) Free model, conventional DoS;
(b) Weak coupling, large coherent peak;
(c) Intermediate coupling, **narrow coherent peak**;
(d) Strong coupling, lower and upper Hubbard bands.

$$\Downarrow$$

Localisation = flattening of bare bands.

Backup 2: details on the HF method

Hartree-Fock method: $\mathcal{S}$ is the set of all **gaussian states**, i.e. states that are non-interacting for some fermionic operators

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle \ , \qquad \left\{\hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger}\right\} = \delta_{\alpha\beta} \ ,$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

$$\Downarrow$$

Wick's theorem: averages of quartic operators decompose nicely,

$$\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\delta} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\gamma} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}} .$$

The EHM hamiltonian is a quartic operator!

The self-consistency mapping

Each $|\Psi\rangle \in \mathcal{S}$ is uniquely defined by a particular choice of the **variational parameters** (HFPs)

$$\Delta^{+-}_{\alpha\beta} = \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \,, \qquad \Delta^{++}_{\alpha\beta} = \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \,.$$

Let $\mathbf{v}$ be the vector that collects all HFPs. We aim to find

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle \ ,$$

with $\mathbf{v}$ such that:

Variational picture

Energy is minimised,

$$\left.\frac{\delta E[\Phi]}{\delta\,|\Phi\rangle}\right|_{|\Phi\rangle = |\Psi[\mathbf{v}]\rangle} = 0\,.$$

Self-consistency picture

A self-consistency problem is satisfied,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}\,,$$

being A a non-linear map **uniquely determined**.

The iterative algorithm (iHF)

The HF solution is the **fixed point** of a non-linear map A,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}\,.$$

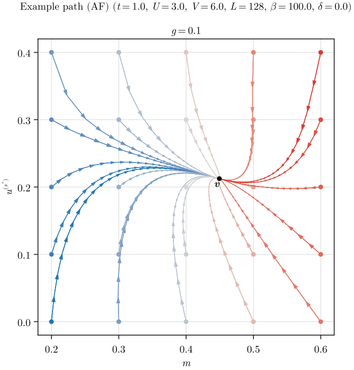
$$\Downarrow$$

Strategy: "follow" the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}(\mathbf{v}_i) + (1-g)\mathbf{v}_i\,.$$

with $i > 0$, $g \in [0, 1]$.

$$\Downarrow$$



Example path for the AF phase.

Convergence criterion:

$$\forall j \quad : \quad [\mathbf{v}_{i+1} - \mathbf{v}_i]_j < \Delta v_j\,,$$

with $\Delta\mathbf{v}$ the vector of tolerances.

Backup 3: Mott localisation in AF and SC phases

Mott localisation in the AF phase

Effective hopping is:

- ⋄ reduced by V;
- ⋄ **increased** by U.

$\Downarrow$

The two interactions "oppose" each other.

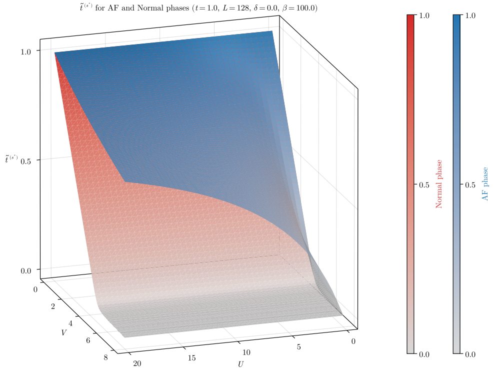


Figure 8: Comparative plot of $\tilde{t}^{(s^*)}$ for normal and AF phase.

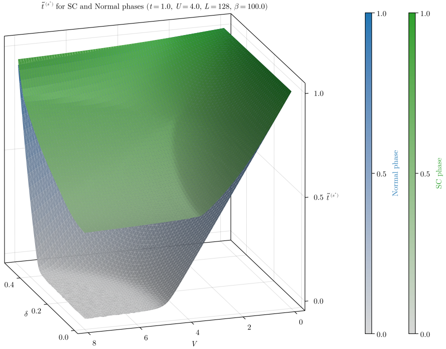


Figure 9: Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the SC phase.

Isotropic hopping renormalisation:

$$\tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)} V}{2} .$$

Formally identical to normal and AF phases.

The effective hopping is mainly isotropic:

$$\left| \tilde{t}^{(s^*)} \right| \gg \left| \tilde{t}^{(d)} \right| .$$

Backup 4: energetic competition between AF and SC phases

The AF-SC phase boundary

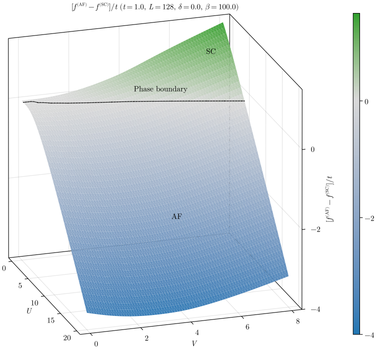


Figure 10: Energy difference: SC versus AF phase.

By computing the free energy difference,

$$f^{(\mathrm{AF})} - f^{(\mathrm{SC})},$$

we identify a phase boundary. Possibly a critical ratio

$$\left(\frac{V}{U}\right)_{\mathrm{c}} = k$$

exists?

The SC phase is dominant when $V \gg U, t$, the AF phase when $U \gg V, t$.